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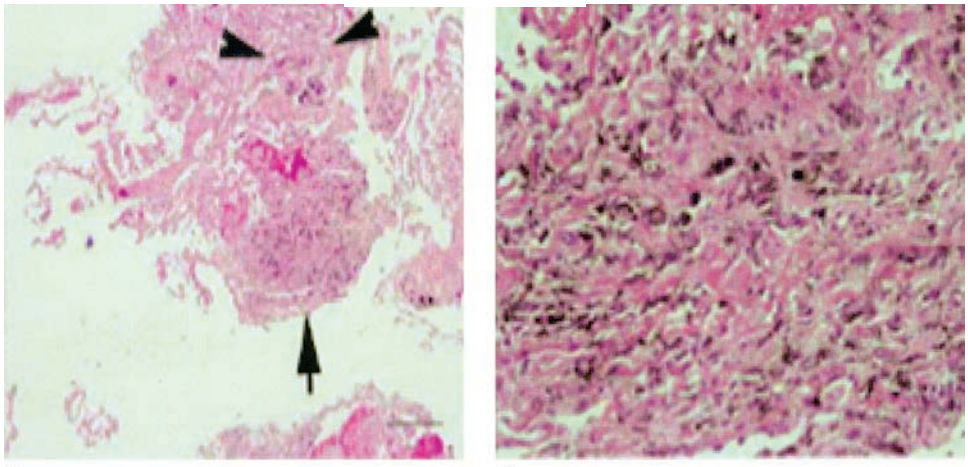


FIGURE 289-4 Histopathologic features of biomass smoke–induced interstitial lung disease. ***A.*** Anthracitic pigment is seen accumulating along alveolar septae (*arrowheads*) and within a pigmented dust macule (*single arrow*). ***B.*** A high-power photomicrograph contains a mixture of fibroblasts and carbon-laden macrophages.

of exposure to biomass smoke, which can include interstitial lung disease (Fig. 289-4). Evidence is beginning to emerge that improved stoves that reduce biomass smoke exposure can reduce risk of respiratory illness in both children and adults.

Household air pollution (HAP) from domestic use of solid fuels also contributes substantially to outdoor air pollution. Contributions from HAP, coal-fired power plants without emission scrubbers, and increased traffic congestion involving motor vehicles without pollution controls can lead to high concentrations of outdoor air pollution, especially fine PM, in mega-cities in developing countries (e.g., Delhi).

Acknowledgment

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FURTHER READING

Almberg KS et al: Progressive massive fibrosis resurgence identified in U.S. coal miners filing for Black Lung Benefits, 1970-2016. Ann Am Thorac Soc 15:1420, 2018.

Balmes JR: Household air pollution from domestic combustion of solid fuels and health. J Allergy Clin Immunol 143:1979, 2019.

Blanc PD et al: The occupational burden of nonmalignant respiratory diseases. An official American Thoracic Society and European Respiratory Society statement. Am J Respir Crit Care Med 199:1312, 2019.

Gauderman WJ et al: Association of improved air quality with lung development in children. N Engl J Med 372:905, 2015.

Leon-Jimenez A et al: Artificial stone silicosis: Rapid progression following exposure cessation. Chest 158:1060, 2020.

Musk AW et al: Asbestos-related diseases. Int J Tuberc Lung Dis 24:562, 2020.

290 Bronchiectasis

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Bronchiectasis refers to an irreversible airway dilation that involves the lung in either a focal or a diffuse manner and that classically has been categorized as cylindrical or tubular (the most common form), varicose, or cystic. This chapter will focus largely on non–cystic fibrosis (CF) bronchiectasis. **The reader is referred to Chapter 291 for a more focused discussion on CF bronchiectasis.**

ETIOLOGY

Bronchiectasis can arise from infectious or noninfectious causes (Table 290-1). Clues to the underlying etiology often are provided by the pattern of lung involvement. *Focal bronchiectasis* refers to bronchiectatic changes in a localized area of the lung and can be a consequence of obstruction of the airway—either extrinsic (e.g., due to compression by adjacent lymphadenopathy or parenchymal tumor mass) or intrinsic (e.g., due to an airway tumor or aspirated foreign body, a scarred/stenotic airway, or bronchial atresia from congenital underdevelopment of the airway). *Diffuse bronchiectasis* is characterized by widespread bronchiectatic changes throughout the lung and often arises from an underlying systemic or infectious disease process.

More pronounced involvement of the upper lung fields is most common in CF and also is observed in postradiation fibrosis, corresponding to the lung region encompassed by the radiation port. Bronchiectasis with predominant involvement of the lower lung fields usually has its source in chronic recurrent aspiration (e.g., due to esophageal motility disorders like those in scleroderma), end-stage fibrotic lung disease (e.g., traction bronchiectasis from idiopathic pulmonary fibrosis), or recurrent immunodeficiency-associated infections (e.g., hypogammaglobulinemia). Bronchiectasis resulting from infection by nontuberculous mycobacteria (NTM), most commonly the *Mycobacterium avium-intracellulare* complex (MAC), often preferentially affects the midlung fields. Congenital causes of bronchiectasis with predominant midlung field involvement include the dyskinetic/immotile cilia syndrome. Finally, predominant involvement of the central airways is reported in association with allergic bronchopulmonary aspergillosis (ABPA), in which an immune-mediated reaction to *Aspergillus* damages the bronchial wall. Congenital causes of central airway–predominant bronchiectasis resulting from cartilage deficiency include tracheobronchomegaly (Mounier-Kuhn syndrome) and Williams-Campbell syndrome.

In many cases, the etiology of bronchiectasis is not determined. In case series, as many as 25–50% of patients referred for bronchiectasis have idiopathic disease.

EPIDEMIOLOGY

The overall reported prevalence of bronchiectasis in the United States has recently increased, but the epidemiology of bronchiectasis varies greatly with the underlying etiology. For example, patients born with CF often develop significant clinical bronchiectasis in late adolescence or early adulthood, although atypical presentations of CF in adults in their thirties and forties are also possible. In contrast, bronchiectasis resulting from MAC infection classically affects nonsmoking women >50 years of age. In general, the incidence of bronchiectasis increases with age. Bronchiectasis is more common among women than among men. Bronchiectasis may also frequently be co-diagnosed with chronic obstructive pulmonary disease (COPD) or asthma.

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TABLE 290-1 Major Etiologies of Bronchiectasis and Proposed Workup

PATTERN OF LUNG INVOLVEMENT	ETIOLOGY BY CATEGORY (EXAMPLES)	WORKUP
Focal	Obstruction (aspirated foreign body, tumor mass)	Chest imaging (chest x-ray and/or chest CT); bronchoscopy
Diffuse	Infection (bacterial, nontuberculous mycobacterial)	Sputum Gram's stain/culture; stains/cultures for acid-fast bacilli and fungi. If no pathogen is identified, consider bronchoscopy with bronchoalveolar lavage.
	Immunodeficiency (hypogammaglobulinemia, HIV infection, bronchiolitis obliterans after lung transplantation)	Complete blood count with differential; immunoglobulin measurement; HIV testing
	Genetic causes (cystic fibrosis, Kartagener's syndrome, α_1 antitrypsin deficiency)	Measurement of chloride levels in sweat (for cystic fibrosis), α_1 antitrypsin levels; nasal or respiratory tract brush/biopsy (for dyskinetic/immotile cilia syndrome); genetic testing
	Autoimmune or rheumatologic causes (rheumatoid arthritis, Sjögren's syndrome, inflammatory bowel disease); immune-mediated disease (allergic bronchopulmonary aspergillosis)	Clinical examination with careful joint exam, serologic testing (e.g., for rheumatoid factor). Consider workup for allergic bronchopulmonary aspergillosis, especially in patients with refractory asthma.[a]
	Recurrent aspiration	Test of swallowing function and general neuromuscular strength
	Miscellaneous (yellow nail syndrome, traction bronchiectasis from postradiation fibrosis or idiopathic pulmonary fibrosis)	Guided by clinical condition
	Idiopathic	Exclusion of other causes

[a]Skin testing for *Aspergillus* reactivity; measurement of serum precipitins for *Aspergillus*, serum IgE levels, serum eosinophils, etc.

In areas where tuberculosis is prevalent, bronchiectasis more frequently occurs as a sequela of granulomatous infection. Focal bronchiectasis can arise from extrinsic compression of the airway by enlarged granulomatous lymph nodes and/or from development of intrinsic obstruction as a result of erosion of a calcified lymph node through the airway wall (e.g., broncholithiasis). Especially in reactivated tuberculosis, parenchymal destruction from infection can result in areas of more diffuse bronchiectasis. Apart from cases associated with tuberculosis, an increased incidence of non-CF bronchiectasis with an unclear underlying mechanism has been reported as a significant problem in developing nations. It has been suggested that the high incidence of malnutrition in certain areas may predispose to immune dysfunction and development of bronchiectasis.

PATHOGENESIS AND PATHOLOGY

The most widely cited mechanism of infectious bronchiectasis is the "vicious cycle hypothesis," in which susceptibility to infection and poor mucociliary clearance result in microbial colonization of the bronchial tree. Some organisms, such as *Pseudomonas aeruginosa*, exhibit a particular propensity for colonizing damaged airways and evading host defense mechanisms. Impaired mucociliary clearance can result from inherited conditions such as CF or dyskinetic cilia syndrome, and it has been proposed that a single severe infection (e.g., pneumonia caused by *Bordetella pertussis* or *Mycoplasma pneumoniae*) can result in significant airway damage and poor secretion clearance. The presence of the microbes incites continued chronic inflammation, with consequent damage to the airway wall, continued impairment of secretions and microbial clearance, and ongoing propagation of the infectious/inflammatory cycle. Moreover, it has been proposed that mediators released directly from bacteria can interfere with mucociliary clearance.

Classic studies of the pathology of bronchiectasis from the 1950s demonstrated significant small-airway wall inflammation and larger-airway wall destruction as well as dilation, with loss of elastin, smooth muscle, and cartilage. It has been proposed that inflammatory cells in the small airways release proteases and other mediators, such as reactive oxygen species and proinflammatory cytokines, that damage the larger airway walls. Furthermore, the ongoing inflammatory process in the smaller airways results in airflow obstruction. It is thought that antiproteases, such as α_1 antitrypsin, play an important role in neutralizing the damaging effects of neutrophil elastase and in enhancing bacterial killing. Bronchiectasis and emphysema have been observed in patients with α_1 antitrypsin deficiency.

Proposed mechanisms for noninfectious bronchiectasis include immune-mediated reactions that damage the bronchial wall (e.g., those associated with systemic autoimmune conditions such as Sjögren's syndrome and rheumatoid arthritis). Recent studies suggest that there might exist a new bronchiectasis endophenotype of patients with sensitization to multiple environmental allergens. *Traction bronchiectasis* refers to dilated airways arising from parenchymal distortion as a result of lung fibrosis (e.g., postradiation fibrosis or idiopathic pulmonary fibrosis).

CLINICAL MANIFESTATIONS

The most common clinical presentation is a persistent productive cough with ongoing production of thick, tenacious sputum. Physical findings frequently include crackles and wheezing on lung auscultation, and some patients with bronchiectasis exhibit clubbing of the digits. Mild to moderate airflow obstruction often is detected on pulmonary function tests, overlapping with that seen at presentation with other conditions, such as COPD. Acute exacerbations of bronchiectasis usually are characterized by changes in the nature of sputum production, with increased volume and purulence. However, typical signs and symptoms of lung infection, such as fever and new infiltrates, may not be present.

DIAGNOSIS

The diagnosis usually is based on presentation with a persistent chronic cough and sputum production accompanied by consistent radiographic features. Although chest radiographs lack sensitivity, the presence of "tram tracks" indicating dilated airways is consistent with bronchiectasis. Chest CT is more specific for bronchiectasis and is the imaging modality of choice for confirming the diagnosis. CT findings include airway dilation (detected as parallel "tram tracks" or as the "signet-ring sign"—a cross-sectional area of the airway with a diameter at least 1.5 times that of the adjacent vessel), lack of bronchial tapering (including the presence of tubular structures within 1 cm from the pleural surface), bronchial wall thickening in dilated airways, inspissated secretions (e.g., the "tree-in-bud" pattern), or cysts emanating from the bronchial wall (especially pronounced in cystic bronchiectasis) **(Fig. 290-1)**.

APPROACH TO THE PATIENT

Bronchiectasis

The evaluation of a patient with bronchiectasis entails elicitation of a clinical history, chest imaging, and a workup to determine the underlying etiology. Evaluation of focal bronchiectasis almost always requires bronchoscopy to exclude airway obstruction by an underlying mass or foreign body. A workup for diffuse bronchiectasis includes analysis for the major etiologies (Table 290-1), with an initial focus on excluding CF. Pulmonary function testing is an important component of a functional assessment of the patient.

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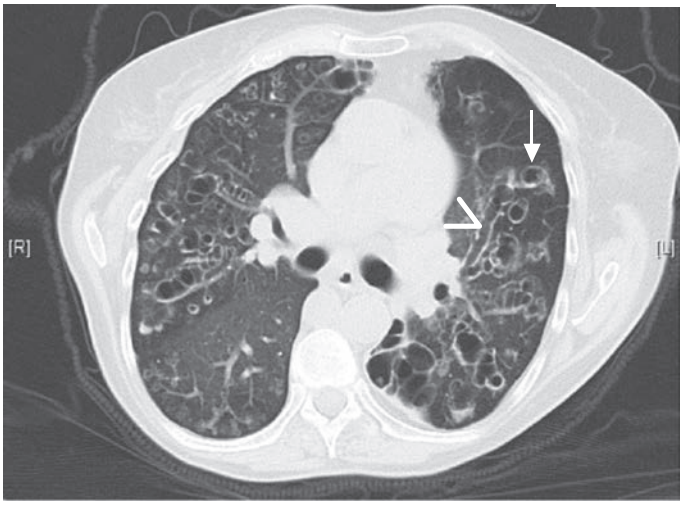


FIGURE 290-1 Representative chest CT image of severe bronchiectasis. This patient's CT demonstrates many severely dilated airways, seen both longitudinally (*arrowhead*) and in cross-section (*arrow*).

TREATMENT

Bronchiectasis

Treatment of infectious bronchiectasis is directed at the control of active infection and improvements in secretion clearance and bronchial hygiene so as to decrease the microbial load within the airways and minimize the risk of repeated infections.

ANTIBIOTIC TREATMENT

Antibiotics targeting the causative or presumptive pathogen (with *Haemophilus influenzae* and *P. aeruginosa* isolated commonly) should be administered in acute exacerbations, usually for a minimum of 7–10 days and perhaps for as long as 14 days. Decisions about treatment of NTM infection can be difficult, given that these organisms can be colonizers as well as pathogens and the prolonged treatment course often is not well tolerated. Consensus guidelines have advised that diagnostic criteria for true clinical infection with NTM should be considered in patients with symptoms and radiographic findings of lung disease who have at least two sputum samples positive on culture; at least one bronchoalveolar lavage (BAL) fluid sample positive on culture; a biopsy sample displaying histopathologic features of NTM infection (e.g., granuloma or a positive stain for acid-fast bacilli) along with one positive sputum culture; or a pleural fluid sample (or a sample from another sterile extrapulmonary site) positive on culture. MAC strains are the most common NTM pathogens, and the recommended regimen for HIV-negative patients infected with macrolide-sensitive MAC includes a macrolide combined with rifampin and ethambutol. Consensus guidelines recommend macrolide susceptibility testing for clinically significant MAC isolates.

BRONCHIAL HYGIENE

The numerous approaches used to enhance secretion clearance in bronchiectasis include hydration and mucolytic administration, aerosolization of bronchodilators and hyperosmolar agents (e.g., hypertonic saline), and chest physiotherapy (e.g., postural drainage, traditional mechanical chest percussion via hand clapping to the chest, or use of devices such as an oscillatory positive expiratory pressure flutter valve or a high-frequency chest wall oscillation vest). Pulmonary rehabilitation and a regular exercise program may assist with secretion clearance as well as with other aspects of bronchiectasis, including improved exercise capacity and quality of life. The mucolytic dornase (DNase) is recommended routinely in CF-related bronchiectasis but not in non-CF bronchiectasis, given concerns about lack of efficacy and potential harm in the non-CF population.

ANTI-INFLAMMATORY THERAPY

It has been proposed that control of the inflammatory response may be of benefit in bronchiectasis, and relatively small-scale trials have yielded evidence of alleviated dyspnea, decreased need for inhaled β-agonists, and reduced sputum production with inhaled glucocorticoids. However, no significant differences in lung function or bronchiectasis exacerbation rates have been observed. Risks of immunosuppression and adrenal suppression must be carefully considered with use of anti-inflammatory therapy in infectious bronchiectasis. Nevertheless, administration of oral/systemic glucocorticoids may be important in treatment of bronchiectasis due to certain etiologies, such as ABPA, or of noninfectious bronchiectasis due to underlying conditions, especially that in which an autoimmune condition is believed to be active (e.g., rheumatoid arthritis or Sjögren's syndrome). Patients with ABPA also may benefit from a prolonged course of treatment with the oral antifungal agent itraconazole.

REFRACTORY CASES

In select cases, surgery can be considered, with resection of a focal area of suppuration. In advanced cases, lung transplantation can be considered.

COMPLICATIONS

In more severe cases of infectious bronchiectasis, recurrent infections and repeated courses of antibiotics can lead to microbial resistance to antibiotics. In certain cases, combinations of antibiotics that have independent toxicity profiles may be necessary to treat resistant organisms.

Recurrent infections can result in injury to superficial mucosal vessels, with bleeding and, in severe cases, life-threatening hemoptysis. Management of massive hemoptysis usually requires intubation to stabilize the patient, identification of the source of bleeding, and protection of the nonbleeding lung. Control of bleeding often necessitates bronchial artery embolization and, in severe cases, surgery.

PROGNOSIS

Outcomes of bronchiectasis can vary widely with the underlying etiology and comorbid conditions and also may be influenced by the frequency of exacerbations and (in infectious cases) the specific pathogens involved (with worse outcomes associated with *P. aeruginosa* colonization). Increasing attention is being given to defining clinical subphenotypes of bronchiectasis in light of heterogeneous clinical, radiographic, and microbial features and to developing screening tools for the assessment of quality of life and disease severity. In one study, the decline of lung function in patients with non-CF bronchiectasis was similar to that in patients with COPD, with the forced expiratory volume in 1 s (FEV_1) declining by 50–55 mL per year as opposed to 20–30 mL per year for healthy controls.

PREVENTION

Reversal of an underlying immunodeficient state (e.g., by administration of gamma globulin for immunoglobulin-deficient patients) and vaccination of patients with chronic respiratory conditions (e.g., influenza and pneumococcal vaccines) can decrease the risk of recurrent infections. Patients who smoke should be counseled about smoking cessation.

After resolution of an acute infection in patients with recurrences (e.g., ≥3 episodes per year), the use of suppressive antibiotics to minimize the microbial load and reduce the frequency of exacerbations has been proposed. Although there is less consensus about this approach in non-CF-associated bronchiectasis than in CF-related bronchiectasis, small studies have supported benefits of selected therapies. Possible suppressive treatments include (1) administration of an oral antibiotic (e.g., ciprofloxacin) daily for 1–2 weeks per month; (2) use of a rotating schedule of oral antibiotics (to minimize the risk of development of drug resistance); (3) administration of a macrolide antibiotic (see below) daily or three times per week (with mechanisms of possible benefit related to non-antimicrobial properties, such as anti-inflammatory effects and reduction of gram-negative bacillary

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biofilms); (4) inhalation of aerosolized antibiotics (e.g., tobramycin inhalation solution) for select patients on a rotating schedule (e.g., 30 days on, 30 days off), with the goal of decreasing the microbial load without eliciting the side effects of systemic drug administration; other studies examining inhaled aztreonam and inhaled ciprofloxacin formulations have shown conflicting results, suggesting there might be subpopulations of patients with bronchiectasis who might benefit from specific therapies; and (5) intermittent administration of IV antibiotics (e.g., "clean-outs") for patients with more severe bronchiectasis and/or resistant pathogens. In relation to macrolide therapy (point 3 above), a number of double-blind, placebo-controlled, randomized trials have been published in non-CF bronchiectasis and support a benefit of long-term macrolides (6–12 months of azithromycin or erythromycin) in decreasing rates of bronchiectasis exacerbation, mucus production, and decline in lung function. However, two of these studies and a meta-analysis also reported increased macrolide resistance in commensal pathogens, dampening enthusiasm for universal use of macrolides in this setting and raising the question of whether there might be select non-CF bronchiectasis patients with higher morbidity for whom benefits of long-term macrolides might outweigh the risks of emergence of antibiotic resistance. In particular, development of macrolide-resistant NTM is a potential concern, making treatment of those pathogens much more difficult. Furthermore, patients with different patterns of microbial colonization may not all experience similar benefits with macrolide therapy. Therefore, before chronic macrolide therapy is considered, it is advisable to rule out NTM infection and carefully consider each patient's scenario closely, obtaining an electrocardiogram to rule out a prolonged QT interval that might place the patient at increased risk of arrhythmias.

In addition, ongoing consistent attention to bronchial hygiene can promote secretion clearance and decrease the microbial load in the airways.

FURTHER READING

Chalmers JD, Chotirmall SH: Bronchiectasis: New therapies and new perspectives. Lancet Respir Med 6:715, 2018.

Henkle E et al: Characteristics and health-care utilization history of patients with bronchiectasis in US Medicare enrollees with prescription drug plans, 2006-2014. Chest 154:1311, 2018.

Mac Aogáin M et al: Distinct "immunoallertypes" of disease and high frequencies of sensitization in non-cystic fibrosis bronchiectasis. Am J Respir Crit Care Med 199:842, 2019.

Wang D et al: Meta-analysis of macrolide maintenance therapy for prevention of disease exacerbations in patients with noncystic fibrosis bronchiectasis. Medicine (Baltimore) 98:e15285, 2019.

291 Cystic Fibrosis

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CLINICAL FEATURES

Cystic fibrosis (CF) is an autosomal recessive exocrinopathy affecting multiple epithelial tissues. The gene product responsible for CF (the cystic fibrosis transmembrane conductance regulator [CFTR]) serves as an anion channel in the apical (luminal) plasma membranes of epithelial cells and regulates volume and composition of exocrine secretion. An increasingly sophisticated understanding of CFTR molecular genetics and membrane protein biochemistry has facilitated CF drug discovery, with a number of recently approved agents that have transformed the clinical outlook for many with the disease.

Respiratory Manifestations The major morbidity and mortality associated with CF is attributable to pulmonary compromise, characterized by copious hyperviscous and adherent secretions that obstruct small and medium-sized airways. CF respiratory secretions are exceedingly difficult to clear, and a complex bacterial flora that includes *Staphylococcus aureus*, *Haemophilus influenzae*, and *Pseudomonas aeruginosa* (among other pathogens, see below) is routinely cultured from CF sputum. Microbiome analysis has identified dozens of other bacterial species in CF lungs, although a relationship of these less well-characterized organisms to disease progression has not been determined. Robust pulmonary inflammation in the setting of inspissated mucus and chronic bacterial infection leads to collateral tissue injury and further aggravates respiratory decline. Organisms such as *P. aeruginosa* exhibit a stereotypic mode of pathogenesis; a sentinel and early colonization event often engenders lifelong pulmonary infection by the same genetic strain. Over a period of many years, *P. aeruginosa* evolves in CF lungs to adopt a mucoid phenotype (attributable to release of alginate exoproduct) that confers selective advantage for the pathogen and poor prognosis for the host.

Pancreatic Findings The complete name of the disease, *cystic fibrosis of the pancreas*, refers to profound tissue destruction of the exocrine pancreas, with fibrotic scarring and/or fatty replacement, cyst formation, loss of acinar tissue, and ablation of normal pancreatic architecture. As in the lung, tenacious exocrine secretions (sometimes termed *concretions*) obstruct pancreatic ducts and impair production and flow of digestive enzymes to the duodenum. The sequelae of exocrine pancreatic insufficiency include chronic malabsorption, poor growth, fat-soluble vitamin insufficiency, high levels of blood immunoreactive trypsinogen (a test used in newborn screening), and loss of pancreatic islet cell mass. CF-related diabetes mellitus is a manifestation in >30% of adults with the disease and likely multifactorial in nature (attributable to progressive destruction/dysfunction of the endocrine pancreas and, in some cases, insulin resistance or other features).

Additional Organ System Damage As in CF lung and pancreas, thick and inspissated secretions compromise numerous exocrine tissues. Obstruction of intrahepatic bile ducts and parenchymal fibrosis are commonly observed in pathologic specimens, with multilobular cirrhosis in 4–15% of patients with CF and significant hepatic insufficiency as a resulting manifestation among many adults. Contents of the intestinal lumen are often difficult to excrete, leading to meconium ileus (a presentation in 10–20% of newborns with CF) or distal intestinal obstructive syndrome in older individuals. Men typically exhibit complete involution of the vas deferens and infertility (despite functioning spermatogenesis), and ~99% of males with CF are infertile. The etiology of this dramatic anatomic defect in the male genitourinary system is not understood but may represent a developmental abnormality secondary to improper epithelial secretion in the vas or associated structures. Males with CF can conceive children through in vitro fertilization. Abnormalities of female reproductive tract secretions are likely contributors to a higher incidence of infertility among women with the disease. Radiographic evidence of sinusitis occurs in most patients with CF and is associated with organisms similar to those recovered from lower airways, suggesting the sinus may serve as a reservoir for bacterial seeding.

PATHOGENESIS

Cystic Fibrosis Transmembrane Conductance Regulator CFTR is an integral membrane protein that functions as an epithelial anion channel. The ~1480-amino-acid molecule encodes a passive conduit for chloride and bicarbonate transport across plasma membranes of epithelial tissues, with direction of ion flow dependent on the electrochemical driving force. Gating of CFTR involves conformational cycling between an open and closed configuration and is augmented by hydrolysis of adenosine triphosphate (ATP). Anion flux mediated by CFTR does not involve active transport against a concentration gradient but utilizes the energy provided from ATP hydrolysis as a central feature of ion channel mechanochemistry and gating.

CFTR is situated in the apical plasma membranes of acinar and other epithelial cells where it regulates the amount and composition